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implicit constant initialization

Table of contents

- implicit constant initialization
 - Changelog
 - Abstract
 - Motivating examples
 - Classes not Having Value Semantics
 - Returned References to Temporaries
 - Proposed Wording
 - In Depth Rationale
 - Why not before
 - Storage Duration
 - Constant Expressions
 - Constant Initialization
 - constant definition in class definitions
 - constant definition in function body
 - constant definition in function parameter and arguments
 - Impact on current proposals
 - p2255r2
 - p2576r0
 - Past
 - n1511
 - n2235
 - Present
 - C Standard Compound Literals

- C++ Standard
- Outstanding Issues
 - P1018R16
 - LWG2432 initializer_list assignability
 - CWG900 Lifetime of temporaries in range-based for
 - CWG1864 List-initialization of array objects
 - CWG2111 Array temporaries in reference binding
 - CWG914 Value-initialization of array types
- Other languages
- Summary
- Frequently Asked Questions
- References

Changelog

R1

- Clarified existing attribution by
 - adding more links to existing references
 - judiciously used boxes to group quoted material together to improve visibility
- Numerous verbiage clarifications
- Greatly reduced the why not before section
- `std::initializer_list` example automatically fixed without using `const`

Abstract

“Lifetime issues with references to temporaries can lead to fatal and subtle runtime errors. This applies to both:” ^[1]

- *“Returned references (for example, when using strings or maps) and”* ^[1:1]
- *“Returned objects that do not have value semantics (for example using `std::string_view`).”* ^[1:2]

This paper proposes the standard adopt existing common practices in order to eliminate dangling in some cases and in many other cases, greatly reduce them.

Motivating Examples

“Let’s motivate the feature for both, classes not having value semantics and references.” [1:3]

“Classes not Having Value Semantics” [1:4]

“C++ allows the definition of classes that do not have value semantics. One famous example is `std::string_view`: The lifetime of a `string_view` object is bound to an underlying string or character sequence.” [1:5]

“Because `string` has an implicit conversion to `string_view`, it is easy to accidentally program a `string_view` to a character sequence that doesn’t exist anymore.” [1:6]

“A trivial example is this:” [1:7]

```
std::string_view sv = "hello world"s; // immediate dangling reference
```

It is clear from this `string_view` example that it dangles because `sv` is a reference and `"hello world"s` is a temporary. **What is being proposed is that same example doesn’t dangle!** If the evaluated constant expression `"hello world"s` had static storage duration just like the string literal `"hello world"` has static storage duration [2] (5.13.5 String literals [lex.string]) then `sv` would be a reference to something that is global and as such would not dangle. This is reasonable based on how programmers reason about constants being immutable variables and temporaries which are known at compile time and do not change for the life of the program. There are a few facts to take note of in the previous example.

- constants, whether `"hello world"` or `"hello world"s`, **are not expected by programmers** to dangle but rather to be immutable and available for the life of the program in other words `const` and static storage duration
- the `constexpr` constructor of `std::string_view` **expects** a `const` argument
- `sv` is constant-initialized [2:1] (7.7 Constant expressions [expr.const])

Dangling *“can occur more indirectly as follows:” [1:8]*

```
std::string operator+ (std::string_view s1, std::string_view s2) {  
    return std::string{s1} + std::string{s2};  
}  
std::string_view sv = "hi";  
sv = sv + sv; // fatal runtime error: sv refers to deleted temporary string
```

The problem here is that the lifetime of the temporary is bound to the statement in which it was created, instead of the block that contains said expression.

Working Draft, Standard for Programming Language C++ [2:2]

“6.7.7 Temporary objects”

“Temporary objects are destroyed as the last step in evaluating the full-expression (6.9.1) that (lexically) contains the point where they were created. This is true even if that evaluation ends in throwing an exception. The value computations and side effects of destroying a temporary object are associated only with the full-expression, not with any specific subexpression.”

Had the temporary been bound to the enclosing block than it would have been alive for at least as long as the returned reference. While this does reduce dangling, it does not eliminate it because if the reference out lives its containing block such as by returning than dangling would still occur. These remaining dangling would at least be more visible as they are usually associated with returns, so you know where to look and if we make the proposed changes than there would be far fewer dangling to look for. It should also be noted that **the current lifetime rules of temporaries are like constants, contrary to programmer’s expectations**. This becomes more apparent with slightly more complicated examples.

“Returned References to Temporaries” [1:9]

“Similar problems already exists with references.” [1:10]

“A trivial example would be the following:” [1:11]

```
struct X { int a, b; };  
int& f(X& x) { return x.a; } // return value lifetime bound to parameter
```

If `f` was called with a temporary than it too would dangle.

```
int& a = f({4, 2});  
a = 5; // fatal runtime error
```

If the lifetime of the temporary, `{4, 2}`, was bound to the lifetime of its containing block instead of its containing statement than `a` would not immediately dangle. Further, `{4, 2}` is constant initialized, so if function `f`’s signature was changed to be `int& f(const X& x)`, since it does not change `x`, then this example would not dangle at all.

“Class `std::string` provides such an interface in the current C++ runtime library. For example:” [1:12]

```
char& c = std::string{"hello my pretty long string"}[0];  
c = 'x'; // fatal runtime error  
std::cout << "c: " << c << '\n'; // fatal runtime error
```

Again, if the lifetime of the temporary, `std::string{"hello my pretty long string"}`, was bound to the lifetime of its containing block instead of its containing statement than `c` would not immediately dangle. Further, this more complicated compound temporary expression better illustrates why the current lifetime rules of temporaries are contrary to programmer’s expectations. First of all, let’s rewrite the example, as a programmer would, adding names to everything unnamed.

```
auto anonymous = std::string{"hello my pretty long string"};  
char& c = anonymous[0];  
c = 'x'; // fatal runtime error  
std::cout << "c: " << c << '\n'; // fatal runtime error
```

Even though, the code is the same from a programmer’s perspective, the latter does not dangle while the former do. **Should just naming temporaries, thus turning them into variables, fix memory issues? Should just leaving variables unnamed as temporaries introduce memory issues?** Again, contrary to programmer’s expectations. If we viewed unnecessary/superfluous/immediate dangling as overhead, then the current rules of temporary and constant initialization could be viewed as violations of the zero-overhead principle since just naming temporaries is reasonably written better by hand.

“There are more tricky cases like this. For example, when using the range-base for loop:” [1:13]

```
for (auto x : reversed(make_vector()))
```

“with one of the following definitions, either:” [1:14]

```
template<Range R>  
reversed_range reversed(R&& r) {  
    return reversed_range{r}; // return value lifetime bound to parameter  
}
```

“or” [1:15]

```
template<Range R>
reversed_range reversed(R r) {
    return reversed_range{r}; // return value lifetime bound to parameter
}
```

Yet again, if the lifetime of the temporary, `reversed(make_vector())`, was bound to the lifetime of its containing block instead of its containing statement than `x` would not immediately dangle. Before adding names to everything unnamed, we must expand the range based for loop.

```
{// containing block
    auto&& rg = reversed(make_vector());
    auto pos = rg.begin();
    auto end = rg.end();
    for ( ; pos != end; ++pos ) {
        auto x = *pos;
        ...
    }
}
```

Now, let's rewrite that expansion, as a programmer would, adding names to everything unnamed.

```
{// containing block
    auto anonymous1 = make_vector();
    auto anonymous2 = reversed(anonymous1);
    auto pos = anonymous2.begin();
    auto end = anonymous2.end();
    for ( ; pos != end; ++pos ) {
        auto x = *pos;
        ...
    }
}
```

Like before, the named version doesn't dangle and as such binding the lifetime of the temporary to the containing block makes more sense to the programmer than binding the lifetime of the temporary to the containing statement. In essence, from a programmer's perspective, **temporaries are anonymously named variables**.

It should be noted too that the current rules of temporaries discourages the use of temporaries because of the dangling it introduces. However, if the lifetime of temporaries was increased to a reasonable degree than programmers would use temporaries more. This would reduce dangling

further because there would be fewer named variables that could be propagated outside of their containing scope. This would also improve code clarity by reducing the number of lines of code allowing any remaining dangling to be more clearly seen.

“Finally, such a feature would also help to ... fix several bugs we see in practice:” [1:16]

“Consider we have a function returning the value of a map element or a default value if no such element exists without copying it.” [1:17]

```
const V& findOrDefault(const std::map<K,V>& m, const K& key, const V& defvalue);
```



“then this results in a classical bug:” [1:18]

```
std::map<std::string, std::string> myMap;  
const std::string& s = findOrDefault(myMap, key, "none"); // runtime bug if key no
```



Is this really a bug? With this proposal, it isn't! Here is why. The function `findOrDefault` **expects** a `const string&` for its third parameter. Since C++20, `string`'s constructor is `constexpr`. It **CAN** be constructed as a constant expression. Since all the arguments passed to this `constexpr` constructor are constant expressions, in this case `"none"`, the temporary `string defvalue` **IS** also constant-initialized [2:3] (7.7 Constant expressions [expr.const]). This paper advises that if you have a non mutable `const` that it is constant-initialized, that the variable or temporary undergoes constant initialization [2:4] (6.9.3.2 Static initialization [basic.start.static]). In other words it has implicit static storage duration. The temporary would actually cease to be a temporary. As such this usage of `findOrDefault` **CAN'T** dangle.

What if `defvalue` can't be constant-initialized because it was created at runtime. If the temporary `string`'s lifetime was bound to the containing block instead of the containing statement than the chance of dangling is greatly reduced and also made more visible. You can say that it **CAN'T** immediately dangle. However, dangling still could occur if the programmer manually propagated the returned value that depends upon the temporary outside of the containing scope.

While using the containing's scope instead of the statement's scope is a vast improvement. We can actually do a little bit better. Following is an example of uninitialized and delayed initialization.

```

bool test();

struct X { int a, b; };

const X* ref2pointer(const X& ref)
{
    return &ref;
}

X x_factory(int a, int b)
{
    return {a, b};
}

int main()
{
    const X* x; // uninitialized
    if(test())
    {
        x = ref2pointer({2, 4}); // implicit constant initialization
    }
    else
    {
        x = ref2pointer(x_factory(4, 2)); // delayed initialization
        // statement scope or
        // containing scope or
        // scope of the variable to which the temporary is assigned
    }
}

```

According to this proposal, `ref2pointer({2, 4})` would undergo implicit constant initialization so that expression would not dangle.

The variable `x` would dangle if initialized with the expression `ref2pointer(x_factory(4, 2))` when the scope is bound to the containing statement. The variable would also dangle if initialized with the expression `ref2pointer(x_factory(4, 2))` when the scope is bound to the containing block. The variable would NOT dangle if initialized with the expression `ref2pointer(x_factory(4, 2))` when the scope is bound to the lifetime of the variable to which the temporary is assigned, in this case `x`.

Extending the lifetime of the temporary to be the lifetime of the variable to which it is assigned is not unreasonable for C++. Matter of fact it is already happening but the rules are so restrictive that it limits its use by many programmers as the following examples illustrate.

“6.7.7 Temporary objects”

...

“₅ There are ... contexts in which temporaries are destroyed at a different point than the end of the full expression.”

...

“(6.8)”

```
template<typename T> using id = T;

int i = 1;
int&& a = id<int[3]>{1, 2, 3}[i]; // temporary array has same lifetime as a
const int& b = static_cast<const int&>(0); // temporary int has same lifetime as b
int&& c = cond ? id<int[3]>{1, 2, 3}[i] : static_cast<int&&>(0);
// exactly one of the two temporaries is lifetime-extended
```



```
const int& x = (const int&)1; // temporary for value 1 has same lifetime as x
```

```
struct S {
    const int& m;
};
const S& s = S{1}; // both S and int temporaries have lifetime of s
```

The preceding sections of this proposal is identical at times in wording, in structure as well as in examples to p0936r0, the Bind Returned/Initialized Objects to the Lifetime of Parameters ^[1:19] proposal. This shows that similar problems can be solved with simpler solutions, that programmers are already familiar with, such as constants and naming temporaries. It must be conceded that Bind Returned/Initialized Objects to the Lifetime of Parameters ^[1:20] is a more general solution that fixes more dangling while this proposal is more easily understood by programmers of all experience levels but fixes less dangling.

Why not just extend the lifetime as prescribed in Bind Returned/Initialized Objects to the Lifetime of Parameters ?

In that proposal, a question was raised.

“Lifetime Extension or Just a Warning?” “We could use the marker in two ways:”

1. *“Warn only about some possible buggy behavior.”*

2. “Fix possible buggy behavior by extending the lifetime of temporaries”

In reality, there are three scenarios; warning, **error** or just fix it by extending the lifetime.

However, things in the real world tend to be more complicated. Depending upon the scenario, at least theoretically, some could be fixed, some could be errors and some could be warnings. Further, waiting on a more complicated solution that can fix everything may never happen or worse be so complicated that the developer no longer can understand the lifetimes of the objects created. Shouldn't we fix what we can, when we can; i.e. low hanging fruit. Also, fixing everything the same way would not even be desirable. Let's consider a real scenario. Extending one's lifetime could mean 2 different things.

1. Change automatic storage duration such that a instances' lifetime is just moved lower on the stack as prescribed in p0936r0.
2. Change automatic storage duration to static storage duration. [This is what I am proposing but only for those that it logically applies to.]

If only #1 was applied holistically via p0936r0, `-wlifetime` or some such, then that would not be appropriate or reasonable for those that really should be fixed by #2. Likewise #2 can't fix all but DOES make sense for those that it applies to. As such, this proposal and p0936r0 ^[1:21] are complimentary.

Personally, p0936r0 ^[1:22] should be adopted regardless because we give the compiler more information than it had before, that argument(s) lifetime is dependent upon the return(s) lifetime. When we give more information, like we do with `const` and `constexpr`, the `c++` compiler can do amazing things. Any reduction in undefined behavior, dangling references/pointers and delayed/uninitialized errors should be welcomed, at least as long it can be explained simply and rationally.

Proposed Wording

6.7.5.4 Automatic storage duration [basic.stc.auto]

¹ Variables that belong to a block or parameter scope and are not explicitly declared `static`, `thread_local`, `extern` or had not underwent implicit constant initialization (6.9.3.2) have automatic storage duration. The storage for these entities lasts until the block in which they are created exits.

...

6.7.7 Temporary objects

...

4 When an implementation introduces a temporary object of a class that has a non-trivial constructor (11.4.5.2, 11.4.5.3), it shall ensure that a constructor is called for the temporary object. Similarly, the destructor shall be called for a temporary with a non-trivial destructor (11.4.7). Temporary objects are destroyed via automatic storage duration (6.7.5.4) as if the compiler was naming the temporaries anonymously or when the variable to which the temporary is assigned is destroyed, whichever is greater lifetime as the last step in evaluating the full-expression (6.9.1) that (lexically) contains the point where they were created. This is true even if that evaluation ends in throwing an exception. The value computations and side effects of destroying a temporary object are associated only with the full expression, not with any specific subexpression.

5 ~~There are three contexts in which temporaries are destroyed at a different point than the end of the full expression. The first context is when a default constructor is called to initialize an element of an array with no corresponding initializer (9.4). The second context is when a copy constructor is called to copy an element of an array while the entire array is copied (7.5.5.3, 11.4.5.3). In either case, if the constructor has one or more default arguments, the destruction of every temporary created in a default argument is sequenced before the construction of the next array element, if any.~~

6 ~~The third context is when a reference binds to a temporary object.²⁹ The temporary object to which the reference is bound or the temporary object that is the complete object of a subobject to which the reference is bound persists for the lifetime of the reference if the glvalue to which the reference is bound was obtained through one of the following:~~

~~(6.1) — a temporary materialization conversion (7.3.5);~~

~~(6.2) — (expression), where expression is one of these expressions;~~

~~(6.3) — subscripting (7.6.1.2) of an array operand, where that operand is one of these expressions;~~

~~(6.4) — a class member access (7.6.1.5) using the . operator where the left operand is one of these expressions and the right operand designates a non-static data member of non-reference type;~~

~~(6.5) — a pointer to member operation (7.6.4) using the .* operator where the left operand is one of these expressions and the right operand is a pointer to data member of non-reference type;~~

~~(6.6) — a~~

~~(6.6.1) — const_cast (7.6.1.11);~~

~~(6.6.2) — static_cast (7.6.1.9);~~

~~(6.6.3) — dynamic_cast (7.6.1.7), or~~

~~(6.6.4) — reinterpret_cast (7.6.1.10)~~

~~converting, without a user-defined conversion, a glvalue operand that is one of these expressions to a glvalue that refers to the object designated by the operand, or to its complete object or a subobject thereof,~~

~~(6.7) — a conditional expression (7.6.16) that is a glvalue where the second or third operand is one of these expressions, or~~

~~(6.8) — a comma expression (7.6.20) that is a glvalue where the right operand is one of these expressions.~~

[Example 2:

```
template<typename T> using id = T;  
  
int i = 1;  
int&& a = id<int[3]>{1, 2, 3}[i]; // temporary array has same lifetime as a  
const int& b = static_cast<const int&>(0); // temporary int has same lifetime as b  
int&& c = cond ? id<int[3]>{1, 2, 3}[i] : static_cast<int&&>(0);  
// exactly one of the two temporaries is lifetime extended
```

~~—end example]~~

~~[Note 5: An explicit type conversion (7.6.1.4, 7.6.3) is interpreted as a sequence of elementary casts, covered above.~~

[Example 3:

```
const int& x = (const int&)1; // temporary for value 1 has same lifetime as x
```

~~—end example]~~

~~—end note]~~

~~[Note 6: If a temporary object has a reference member initialized by another temporary object, lifetime extension applies recursively to such a member's initializer.~~

[Example 4:

```

struct S {
    const int& m;
};
const S& s = S{1}; // both S and int temporaries have lifetime of s

```

~~—end example}~~

~~—end note}~~

The exceptions to this lifetime rule are:

~~(6.9) — A temporary object bound to a reference parameter in a function call (7.6.1.3) persists until the completion of the full expression containing the call.~~

~~(6.10) — A temporary object bound to a reference element of an aggregate of class type initialized from a parenthesized expression list (9.4) persists until the completion of the full expression containing the expression list.~~

~~(6.11) — The lifetime of a temporary bound to the returned value in a function return statement (8.7.4) is not extended; the temporary is destroyed at the end of the full expression in the return statement.~~

~~(6.12) — A temporary bound to a reference in a new initializer (7.6.2.8) persists until the completion of the full expression containing the new initializer.~~

~~{Note 7: This might introduce a dangling reference. —end note}~~

~~{Example 5:~~

```

struct S { int mi; const std::pair<int,int>& mp; };
S a { 1, {2,3} };
S* p = new S{ 1, {2,3} }; // creates dangling reference

```

~~—end example}~~

~~z The destruction of a temporary whose lifetime is not extended by being bound to a reference is sequenced before the destruction of every temporary which is constructed earlier in the same full expression. If the lifetime of two or more temporaries to which references are bound ends at the same point, these temporaries are destroyed at that point in the reverse order of the completion of their construction. In addition, the destruction of temporaries bound to references shall take into account the ordering of destruction of objects with static, thread, or automatic storage duration (6.7.5.2, 6.7.5.3, 6.7.5.4); that is, if obj1 is an object with the same storage duration as the temporary and created before the temporary is created the temporary shall be~~

destroyed before obj1 is destroyed; if obj2 is an object with the same storage duration as the temporary and created after the temporary is created the temporary shall be destroyed after obj2 is destroyed.

g[Example 6:

```
struct S {  
    S();  
    S(int);  
    friend S operator+(const S&, const S&);  
    ~S();  
};  
S obj1;  
const S& cr = S(16)+S(23);  
S obj2;
```

The expression `S(16) + S(23)` creates three temporaries: a first temporary T1 to hold the result of the expression `S(16)`, a second temporary T2 to hold the result of the expression `S(23)`, and a third temporary T3 to hold the result of the addition of these two expressions. The temporary T3 is then bound to the reference `cr`. It is unspecified whether T1 or T2 is created first. On an implementation where T1 is created before T2, T2 shall be destroyed before T1. The temporaries T1 and T2 are bound to the reference parameters of `operator+`; these temporaries are destroyed at the end of the full expression containing the call to `operator+`. The temporary T3 bound to the reference `cr` is destroyed at the end of `cr`'s lifetime, that is, at the end of the program. In addition, the order in which T3 is destroyed takes into account the destruction order of other objects with static storage duration. That is, because `obj1` is constructed before T3, and T3 is constructed before `obj2`, `obj2` shall be destroyed before T3, and T3 shall be destroyed before `obj1`.

—end example]

6.9.3.2 Static initialization [basic.start.static]

...

2 Constant initialization is performed explicitly if a variable or temporary object with static or thread storage duration is constant-initialized (7.7). Constant initialization is performed implicitly if a non mutable const variable or non mutable const temporary object is constant-initialized (7.7). If constant initialization is not performed, a variable with static storage duration (6.7.5.2) or thread storage duration (6.7.5.3) is zero-initialized (9.4). Together, zero-initialization and constant initialization are called static initialization; all other initialization is dynamic initialization. All static initialization strongly happens before (6.9.2.2) any dynamic initialization.

In Depth Rationale

There is a general expectation across programming languages that constants or more specifically constant literals are “immutable values which are known at compile time and do not change for the life of the program”. ^[3] In most programming languages or rather the most widely used programming languages, constants do not dangle. Constants are so simple, so trivial (English wise), that it is shocking to even have to be conscience of dangling. This is shocking to `c++` beginners, expert programmers from other programming languages who come over to `c++` and at times even shocking to experienced `c++` programmers.

Why not before

There is greater need now that more types are getting `constexpr` constructors. Also types that would normally only be dynamically allocated, such as `string` and `vector`, since `c++20`, can also be `constexpr`. This has opened up the door wide for many more types being constructed at compile time.

Storage Duration

Working Draft, Standard for Programming Language C++ ^[2:6]

“5.13.5 String literals *[lex.string]*”

“⁹ **Evaluating a string-literal results in a string literal object with static storage duration** (6.7.5). *Whether all string-literals are distinct (that is, are stored in nonoverlapping objects) and whether successive evaluations of a string-literal yield the same or a different object is unspecified.*”

“*[Note 4: The effect of attempting to modify a string literal object is undefined. — end note]*”

String literals have static storage duration, and thus exist in memory for the life of the program. All the other types of literals have automatic storage duration by default. While that may makes perfect sense for literals that are not constant, constants on other hand are generally believed to be the same value for the life of the program and are ideal candidates to have static storage duration so they can exist in memory for the life of the program. Since literals currently don't behave this way, constant [like] literals are not as simple as they could be, leading to superfluous dangling as well as language inconsistencies.

non dangling	dangling
<pre>const char* non_dangling_42() { const char* constant = "42"; return constant; }</pre>	<pre>const int& dangl { const int co //static con return const }</pre>
<pre>const char* maybe_dangling_42(const char* maybe_constant) { return maybe_constant; } maybe_dangling_42("42");// NOT dangling</pre>	<pre>const int& maybe { return maybe } // dangling beca // 42 is auto no // 42's lifetime maybe_dangling_4</pre>

Even though the usage is the same, the constant string examples do not dangle because they have static storage duration while the dangling, non string literal has automatic storage duration.

Constant Expressions

C++ Core Guidelines: Programming at Compile Time with constexpr ^[4]

“A constant expression”

- *“**can** be evaluated at **compile time**.”*
- *“give the compiler deep insight into the code.”*
- *“are implicitly thread-safe.”*
- *“**can** be constructed in the **read-only memory (ROM-able)**.”*

It isn't just that resolved constant expressions **can** be placed in ROM which makes programmers believe these **should** be stored globally but also the fact that fundamentally **these expressions are executed at compile time**. Along with templates, constant expressions are the closest thing C++ has to **pure** functions. That means the results are the same given the

parameters, and since these expressions run at compile time, than the resultant values are the same, no matter where or when in the `C++` program. This is essentially global to the program; technically across programs too.

This proposal just requests, at least in specific scenarios, that instead of resolved constant-initialized constant expressions **CAN** be ROMable but rather that they **HAVE** to be or at least the next closest thing; constant and `static` storage duration .

Constant Initialization

Working Draft, Standard for Programming Language C++ [2:7]

“6.9.3.2 Static initialization [basic.start.static]”

“₁ Variables with static storage duration are initialized as a consequence of program initiation. Variables with thread storage duration are initialized as a consequence of thread execution. Within each of these phases of initiation, initialization occurs as follows.”

“₂ Constant initialization is performed if a variable or temporary object with static or thread storage duration is constant-initialized (7.7).”

The syntax for `constant` initialization is as follows:

```
static T & ref = constexpr; // constant-initialized required
```

```
static T object = constexpr; // constant-initialized required
```

For the sake of this proposal, I am currently only talking about a subset of `constant` initialization where `T` is `const` since the primary focus is on constants.

```
static const T & ref = constexpr; // constant-initialized required
```

```
static const T object = constexpr; // constant-initialized required
```

“6.7.5.2 Static storage duration [basic.stc.static]”

“All variables which”

- *“do not have thread storage duration and”*
- *“belong to a **namespace scope** (6.4.5) or are first declared with the static or extern keywords (9.2.2) have static storage duration. The storage for these entities lasts for the duration of the program (6.9.3.2, 6.9.3.4).”*

Ironically, at namespace scope, variables are already implicitly static and as such the previous example could simply be written as the following:

```
const T & ref = constexpr; // constant-initialized required
```

```
const T object = constexpr; // constant-initialized required
```

Unfortunately, the static storage duration doesn't come from the fact that it is a constant-initialized constant expression and that a constant was expected/requested. Consider for a moment, if it did, that is implicit static storage duration by looking at this from the perspective of constant definitions in class definition, function body and parameters/arguments.

constant definition in class definitions

```
struct S
{
    const T & ref = constexpr;
    const T object = constexpr;
};
```

If the class members `ref` and `object` were implicitly made static the result to class `S` would be a reduction in size and initialization time. The class members `ref` and `object` would still be `const` and their respective types, so their should be no change in the logic of the code that directly depends upon them. While supporting implicit static storage duration here does not decrease dangling, it would be of benefit from a language simply standpoint for this to be consistent with the next two sections; function body and their parameters/arguments. However since this section has to do with class data member definition and not code execution, explicit static could still be required, if their is a real and significant concern over code breakage.

constant definition in function body

```
const int& dangling_42()
{
    const int& constant = 42;
    return constant;
}
```

```
const int& dangling_42()
{
    const int constant = 42;
    return constant;
}
```

```
const int& dangling_42()
{
    return 42;
}
```

With implicit static storage duration, these examples would not dangle and would make more sense. The local variable `constant` and even unnamed variable, temporary, `42`, still is `const` and have their respective types, so there should be no change in the logic of the code that directly depends upon them. As such, this feature isn't expected to break existing code.

constant definition in function parameter and arguments

```
const int& maybe_dangling_42(const int& maybe_constant);

maybe_dangling_42(42);
```

This is expected to be a non-breaking change because a parameter that takes a `const&` argument doesn't know whether the instance was created locally, statically or even dynamically. The advantage of adding implicit constant initialization here would be that those temporary `const` argument scenarios would no longer dangle. Static storage duration for arguments is actually the primary goal of this paper. It should also be noted that while `static` can be applied explicitly in class data member definition and in function bodies, `static` isn't even an option as a modifier to a function argument, so the user doesn't have a choice and the current default of automatic storage duration instead of static storage duration is less intuitive when constants or constant expressions are involved. Even if the keyword `static` could be applied to arguments, programmer's code will be littered with that keyword as they will start applying it to every argument initialized in a constant expression fashion.

Impact on current proposals

p2255r2

A type trait to detect reference binding to temporary ^[5]

Following is a slightly modified `constexpr` example taken from the p2255r2 ^[5:1] proposal. Only the suffix `s` has been added. It is followed by a non `constexpr` example. Currently, such an example is immediately dangling. Via p2255r2 ^[5:2], both examples become ill formed. However, with this proposal the `constexpr` example becomes valid.

	constexpr	runtime
Examples	<pre>std::tuple<const std::string&> x("hello"s);</pre>	<pre>std::tuple<const s</pre>
Before	<pre>// dangling</pre>	<pre>// dangling</pre>
p2255r2 ^[5:3]	<pre>// ill-formed</pre>	<pre>// ill-formed</pre>
p2255r2 ^[5:4] and this proposal	<pre>// correct</pre>	<pre>// ill-formed</pre>

The proposed valid example is reasonable from many programmers perspective because `"hello"s` is a literal just like `"hello"` is a safe literal in C++. Implicit constant initialization turns constant temporaries into just global variables as such they are no longer temporaries and such would not impact this proposal.

C99 compound literals are **safer** literals because the lifetime is the life of the block containing the temporary instead of the expression. As such turning temporaries into anonymously named variables would remove the need to use this trait to prevent dangling since their would be valid mutable usage of temporaries that no longer immediately dangle.

p2576r0

The constexpr specifier for object definitions ^[6]

The p2576r0 ^[6:1] proposal is about contributing `constexpr` back to the `c` programming language. Interestingly, `c++` has `constexpr` in the first place, in part, to allow `c99` compound literals in `c++`. In the p2576r0 ^[6:2] proposal there are numerous references to “constant expression” and “static storage duration” highlighting that this and my proposal are playing in the same playground. Consider the following:

“C requires that objects with static storage duration are only initialized with constant expressions.

“Because C limits initialization of objects with static storage duration to constant expressions, it can be difficult to create clean abstractions for complicated value generation.”

Further, the p2576r0 ^[6:3] proposal has a whole section devoted to just storage duration.

“3.4. Storage duration”

“For the storage duration of the created objects we go with C++ for compatibility, that is per default we have automatic in block scope and static in file scope. The default for block scope can be overwritten by static or refined by register. It would perhaps be more natural for named constants”

- *“to be addressless (similar to a register declaration or an enumeration),”*
- *“to have static storage duration (imply static even in block scope), or”*
- *“to have no linkage (similar to typedef or block local static)”*

“but we decided to go with C++’s choices for compatibility.”

My proposal would constitute a delay in the p2576r0 ^[6:4] proposal as I am advocating for refining the `c++` choices before contributing `constexpr` back to `c`. I also believe that the p2576r0 ^[6:5] proposal fails to consider a fourth alternative with respect to storage duration and that is to go with how `c` handles compound literals, improve upon it and have `c++` to conform with it. That scenario is the second feature proposed by this proposal.

Past

A brief consideration of the proposals that led to `constexpr` landing as a feature in `c++11` bears weight on the justification of refining constant initialization.

2003

*“This note proposes a notion of **user-defined literals** based on literal constructors without requiring new syntax. If combined with the separate proposal for generalized initializer lists, it becomes a generalization of the **C99 notion of compound literals**.”*

*“However, a constructor is a very general construct and there have been many requests for a way to express literals for user-defined types in such a way that a programmer can be **confident that a value will be constructed at compile time and potentially stored in ROM**. For example:”*

```
complex z(1,2); // the variable z can be constructed at compile time
const complex cz(1,2); // the const cz can potentially be put in ROM
```

*“**Personally, I prefer (1): basically, a value is a literal if it is composed out of literals and implemented by a literal constructor. The problem with that is that some people will not trust compilers to do proper resolution, placement in ROM, placement in text segment, etc. Choosing that solution would require text in the standard to constrain and/or guide implementations.**”*

*“**C99 compound literals**”*

“In C99, it is explicitly allowed to take the address of a compound literal. For example:”

```
f(&(struct foo) { 1,2 });
```

*“**This makes sense only if we assume that the {1,2} is stored in a data segment (like a string literal, but different from a int literal). I see no problem allowing that, as long as it is understood that unless & is explicitly used or the literal, a user-defined literal is an rvalue with which the optimizer has a free hand.**”*

*“**It would be tempting to expand this rule to user-defined literals bound to references. For example:”***

```
void f(complex&);
// ...
f(complex(1,2)); // ok?
```

“However, this would touch upon some rather brittle parts of the overload resolution rules to do with rvalue vs. lvalue. For example:”

```
vector<int> v;  
// ...  
vector<int>().swap(v); // ok  
swap(vector<int>(), v); // would become ok
```

*“I suggest we don’t touch this **unless we are looking at the rvalue/lvalue rules for other reasons.**”*

The other reason why we should re-evaluate user-defined literals bound to references is to reduce dangling references and even dangling pointers. It is also surprising that user defined literals do not work simply without memory issues and that they currently work better in C and practically in every other language than it does in C++. ROM, read only memory, is effectively global/static storage duration and const. Note too that the original motivation for constant expressions are for literals. So constant expressions are effectively literals. Note also the following example provided in that proposal.

```
complex z(1,2); // the variable z can be constructed at compile time  
const complex cz(1,2); // the const cz can potentially be put in ROM
```

While both constant-initialized constant expressions are the same, the detail that decides whether it is a literal that can even be placed in ROM was the `const` keyword. So, `const` constant-initialized constant expressions are constant literals or simply, constants.

n2235

2007

“This paper generalizes the notion of constant expressions to include constant-expression functions and user-defined literals”

*“The goal is ... to **increase C99 compatibility.**”*

“This paper generalizes the notion of constant expressions to include calls to “sufficiently simple” functions (constexpr functions) and objects of user-defined types constructed from “sufficiently simple” constructors (constexpr constructors.)”

*“**simplify the language** definition in the area of constant expression **to match existing practice**”*

“Any enhancement of the notion of constant expressions has to carefully consider the entanglement of many different notions, but strongly related. Indeed, the notion of constant expression appears in different contexts:”

“3. Static initialization of objects with static storage.”

*“Similarly, we do not propose to change the already complex and subtle distinction between “static initialization” and “dynamic initialization”. However **we strive for more uniform and consistency among related C++ language features and compatibility**”*

“3 Problems”

“Most of the problems addressed by this proposal have been discussed in previous papers, especially the initial proposal for Generalized Constant Expressions [DR03], the proposal for Literals for user-defined types [Str03], Generalized initializer lists [DRS03], Initializer lists [SDR05]. What follows is a brief summary.”

“3.4 Unexpected dynamic initialization”

*“**However, it is possible to be surprised by expressions that (to someone) “look const” but are not.**”*

“3.5 Complex rules for simple things”

*“The focus of this proposal is to address the issues mentioned in preceding sections. However, discussions in the Core Working Group at the Berlin meeting (April 2006) concluded that the current rules for integral constant expressions are too complicated, and source of several Defect Reports. **Consequently, a “cleanup”, i.e. adoption of simpler, more general rules is suggested.**”*

“4 Suggestions for C++0x”

“Second, we introduce “literals for user-defined type” based on the notion of constant expression constructors.”

“4.2 Constant-expression data”

“A constant-expression value is a variable or data member declared with the constexpr specifier.”

“As for other const variables, storage need not be allocated for a constant expression datum, unless its address is taken.”

```
// the &x forces x into memory
```

“When the initializer for an ordinary variable (i.e. not a constexpr) happens to be a constant, the compiler can choose to do dynamic or static initialization (as ever).”

“Declaring a constructor constexpr will help compilers to identify static initialization and perform appropriate optimizations (like putting literals in read-only memory.) Note that since “ROM” isn’t a concept of the C++ Standard and what to put into ROM is often a quite subtle design decision, this proposal simply allows the programmer to indicate what might be put into ROM (constant-expression data) rather than trying to specify what actually goes into ROM in a particular implementation.”

“We do not propose to make constexpr a storage-class-specifier because it can be combined with either static or extern or register, much like const.”

Most of these references are given to give us a better idea of the current behavior of constexpr. However, it should be noted that the motivations of this proposal is similar to the motivations for constexpr itself. Mainly ...

- Increase C99 compatibility
- Simplify the language to match existing practice
- Lessen the surprise of unreasonable lifetimes by expressions that look and are const
- Consequently, a “cleanup”, i.e. adoption of simpler, more general rules

Present

This proposal should also be considered in the light of the current standards. A better idea of our current rules is necessary to understanding how they may be simplified for the betterment of C++ .

C Standard Compound Literals

Let's first look at how literals specifically compound literals behave in C . There is still a gap between C99 and C++ and closing or reducing that gap would not only increase our compatibility but also reduce dangling.

2021/10/18 Meneide, C Working Draft [9]

"6.5.2.5 Compound literals"

paragraph 5

*"The value of the compound literal is that of an unnamed object initialized by the initializer list. If the compound literal occurs outside the body of a function, the object has static storage duration; otherwise, it has **automatic storage duration associated with the enclosing block.**"*

The lifetime of this "enclosing block" is longer than that of C++ . In C++ under 6.7.7 Temporary objects [class.temporary] specifically 6.12 states a *temporary bound to a reference in a new-initializer (7.6.2.8) persists until the completion of the full-expression containing the new-initializer.*

gcc [10] describes the result of this gap.

"In C, a compound literal designates an unnamed object with static or automatic storage duration. In C++, a compound literal designates a temporary object that only lives until the end of its full-expression. As a result, well-defined C code that takes the address of a subobject of a compound literal can be undefined in C++, so G++ rejects the conversion of a temporary array to a pointer."

Simply put C has fewer dangling than C++ ! What is more is that C's solution covers both const and non const temporaries! Even though it is C , it is more like C++ than what people give this feature credit for because it is tied to blocks/braces, just like RAIL. This adds more weight that the C way is more intuitive. Consequently, the remaining dangling should be easier to spot for developers not having to look at superfluous dangling.

GCC even takes this a step forward which is closer to what this proposal is advocating. The last reference also says the following.

“As a GNU extension, GCC allows initialization of objects with static storage duration by compound literals (which is not possible in ISO C99 because the initializer is not a constant). It is handled as if the object were initialized only with the brace-enclosed list if the types of the compound literal and the object match. The elements of the compound literal must be constant. If the object being initialized has array type of unknown size, the size is determined by the size of the compound literal.”

Even the `c++` standard recognized that there are other opportunities for constant initialization.

Working Draft, Standard for Programming Language C++ [2:9]

“6.9.3.2 Static initialization [basic.start.static]”

“3 An implementation is permitted to perform the initialization of a variable with static or thread storage duration as a static initialization even if such initialization is not required to be done statically, provided that”

“(3.1) — the dynamic version of the initialization does not change the value of any other object of static or thread storage duration prior to its initialization, and”

“(3.2) — the static version of the initialization produces the same value in the initialized variable as would be produced by the dynamic initialization if all variables not required to be initialized statically were initialized dynamically.”

This proposal is one such opportunity. Besides improving constant initialization, we'll be increasing memory safety by reducing dangling.

Should `c++` just adopt `c99` literal lifetimes being scoped to the enclosing block instead of to the `c++` statement, in lieu of this proposal?

NO, there is still the expectation among programmers that constants, `const` evaluations of `const`-initialized constant expressions, are of static storage duration.

Should `c++` adopt `c99` literal lifetimes being scoped to the enclosing block instead of to the `c++` statement, in addition to this proposal?

YES, `c99` literal lifetimes does not guarantee any reduction in dangling, it just reduces it. This proposal does guarantee but only for `const` evaluations of constant-initialized constant expressions. Combined they would be an even greater reduction in dangling. As such this

proposal and c99 compound literals are complimentary. The remainder can be mitigated by other measures.

Should c++ adopt c99 literal lifetimes being scoped to the enclosing block instead of to the c++ statement?

YES, the c++ standard is currently telling programmers that the first two examples in the following table are equivalent with respect to the lifetime of the temporary {1, 2} and the named variable cz . This is because the lifetime of the temporary {1, 2} is bound to the statement, which means it is destroyed before some_code_after is called.

Given

```
void any_function(const complex& cz);
```

Programmer code	What c++ is actually doing!	Programme
<pre>void main() { some_code_before(); any_function({1, 2}); some_code_after(); }</pre>	<pre>void main() { some_code_before(); { const complex anonymous{1, 2}; any_function(anonymous); } some_code_after(); }</pre>	<pre>void main { some_cc const c any_fun some_cc }</pre>

This is contrary to general programmer expectations and how it behaves in c99 . Besides the fact that a large portion of the c++ community has their start in c and besides the fact that no one, in their right mind, would ever litter their code with superfluous braces for every variable that they would like to be a temporary, there is a more fundamental reason why it is contrary to general programmer expectations. It can actually be impossible to write it that way. Consider another example, now with a return value in which the type does not have a default constructor.

Given

```
no_default_constructor any_function(const complex& cz);
```

Programmer code	<pre> void main() { some_code_before(); no_default_constructor ndc = any_function({1, 2}); some_code_after(ndc); } </pre>
What is c++ doing?	<pre> void main() { some_code_before(); { const complex anonymous{1, 2}; no_default_constructor ndc = any_function(anonymous); } some_code_after(ndc); } </pre>
What is c++ doing?	<pre> void main() { some_code_before(); no_default_constructor ndc; { const complex anonymous{1, 2}; ndc = any_function(anonymous); } some_code_after(ndc); } </pre>

It should be noted that neither of the “ what is C++ doing? ” examples even compile. The first because the variable `ndc` is not accessible to the functional call `some_code_after` . The second because the class `no_default_constructor` doesn’t have a default constructor and as such does not have a uninitialized state. In short, the current `c++` behavior of statement scoping of temporaries instead of containing block scoping is more difficult to reason about because the equivalent code cannot be written by the programmer. As such the `c99` way is simpler, safer and more reasonable.

Regardless, dangling would still be possible, especially for non constant expressions. Those could be fixed by some future, non constant expression proposals.

For instance, stackoverflow has a good example of dangling with `Compound literals storage duration in C` ^[11].

```
/* Example 2 - if statements with braces */

double *coefficients, value;

if(x){
    coefficients = (double[3]) {1.5, -3.0, 6.0};
}else{
    coefficients = (double[3]) {4.5, 1.0, -3.5};
}
value = evaluate_polynomial(coefficients);
```

If `coefficients` was `const`, as it should be since all control paths lead to constant expressions, then even this example would not dangle with this proposal. While this example is an example of dangling, it is also an example of uninitialized or more specifically delayed initialization. Interestingly, perhaps in the future, the binding block in question could be the block where `coefficients` is defined, that is the block containing the uninitialized variable that is assigned to a constant expression, instead of the block where the uninitialized `coefficients` is initiated. This refinement to the `C` and `C++` rules would fix even more non `constexpr` dangling in a simple reasonable way.

Once these trivial dangling is removed from the language, the remaining non `const` dangling could be handled by `Bind Returned/Initialized Objects to the Lifetime of Parameters` ^[1:23] or by some similar proposal preferably by fixing it if it makes sense or by a hard error if it really is a decision that the programmer must make.

C++ Standard

It is also good to consider how the `C++` standard impacts this proposal and how the standard may be impacted by such a proposal.

Working Draft, Standard for Programming Language C++ ^[2:10]

String literals are traditionally one of the most common literals and in `C++` they have static storage duration. This is also the case in many programming languages. As such, these facts leads developers to incorrectly believe that all literals or at least all constant literals have static storage duration.

“5.13.5 String literals [lex.string]”

“9 Evaluating a string-literal results in a string literal object with **static storage duration** (6.7.5). Whether all string-literals are distinct (that is, are stored in nonoverlapping objects) and whether successive evaluations of a string-literal yield the same or a different object is unspecified.”

“[Note 4: **The effect of attempting to modify a string literal object is undefined.** — end note]”

This proposal aligns or adjusts [constant] literals not only with `c` compound literals but also with `c++` string literals. It too should be noted that `c++` is seemingly inconsistent on whether other string like literals should behave lifetime wise like `string`. Are array literals of `static storage duration` ? What about if the array was of characters? What about string like literals such as `std::string` and `std::string_view`?

`c++` also says the “*effect of attempting to modify a string literal object is undefined.*” With us having `const` for so long, there is few reasons left for this to go undefined. Undefined behavior doesn’t make constants and non constant literals any easier to deal with. A string literal could have **static storage duration** for constant expressions and **automatic storage duration** for **non** constant expressions, just like other literals. The lifetime of the `automatic storage duration` could be the `c` rule of the enclosing block since it is safer than `c++` . This would further increase the consistency between string literals and custom/constexpr literals. However, considering that string literals currently have `static storage duration` and we want to reduce dangling instead of increasing it by making the lifetime too narrow, it would be reasonable to include rules for uninitialized and general lifetime extension via `Bind Returned/Initialized Objects to the Lifetime of Parameters` ^[1:24] before nudging string literals closer to non string literals.

The section of the `c++` standard on temporary objects has an example of dangling.

“6.7.7 Temporary objects [class.temporary]”

“(6.12) — A temporary bound to a reference in a new-initializer (7.6.2.8) persists until the completion of the full-expression containing the new-initializer.”

“[Note 7: This might introduce a dangling reference. — end note]”

“[Example 5:”

```
struct S { int mi; const std::pair<int,int>& mp; };  
S a { 1, {2,3} };  
S* p = new S{ 1, {2,3} }; // creates dangling reference
```

“— end example]”

It should be noted that this example is not an example of dangling in `c99` or with my proposal because the temporaries would live as long as their containing block which is sufficient for `a` and `p` in this limited example. Further, since `mp` expects a constant, `std::pair` can be constructed at compile time and was constant-initialized than implicit constant initialization could also occur.

“9.4 Initializers [dcl.init]”

Similarly, the section of the `c++` standard on initializer has multiple examples of dangling.

“9.4.1 General [dcl.init.general]”

```
A a1{1, f()}; // OK, lifetime is extended  
A a2(1, f()); // well-formed, but dangling reference  
A a3{1.0, 1}; // error: narrowing conversion  
A a4(1.0, 1); // well-formed, but dangling reference  
A a5(1.0, std::move(n)); // OK
```

Provided that `A` has a `constexpr` constructor and the second parameter was `const`, with this proposal, `a2` would not be dangling if `f()` too was a constant expression. Also `a4` would also not be dangling. The `a4` example does not need jumping to the signature of the `f` function to figure out if it is a constant expression as is the case of the `a2` example. Really the `a4` example is surprising to current developers that it would be dangling since the native literals `1.0` and `1` can be constant expressions.

It should also be noted that the `c++11` brace initialization does not or should not create another block scope.

“9.4.5 List-initialization [dcl.init.list]”

Similarly, the section of the `c++` standard on list-initialization has an example of dangling.

```
struct A {  
    std::initializer_list<int> i4;  
    A() : i4{ 1, 2, 3 } {} // ill-formed, would create a dangling reference  
};
```

According to this proposal, this example would neither be ill formed or dangling as initializer lists expects a const array, arrays can be constexpr constructed and the array was constant-initialized.

“11.9.6 Copy/move elision [class.copy.elision]”

Similarly, the section of the `c++` standard on copy and move elision has examples of dangling.

```
constexpr A a; // well-formed, a.p points to a  
constexpr A b = g(); // error: b.p would be dangling (7.7)  
void h() {  
    A c = g(); // well-formed, c.p can point to c or be dangling  
}
```

Not only would `b` no longer dangle but the ambiguity of `c.p` could “point to c or be dangling” would be gone.

Outstanding Issues

P1018R16

The `C++ Language Evolution` status pandemic edition ^[12] list some issues that could be effected positively by this proposal. Many of these have the `NAD`, Not A Defect, designation. While these may not presently have been a defect, they were surprising enough to have been brought forth in the first place.

LWG2432 initializer_list assignability

```
auto il1 = {1,2,3};
```

```
il1 = {7,8,9}; // currently well-formed but dangles immediately; should be ill-formed
```

With implicit constant initialization and if `il1` was `const auto` instead of `auto`, this example would not dangle and as such should not be ill-formed. With C99 literal enclosing block lifetime, this example, AS IS, would not dangle.

CWG900 Lifetime of temporaries in range-based for

```
// some function
```

```
std::vector<int> foo();
```

```
// correct usage
```

```
auto v = foo();
```

```
for( auto i : reverse(v) ) { std::cout << i << std::endl; }
```

```
// problematic usage
```

```
for( auto i : reverse(foo()) ) { std::cout << i << std::endl; }
```

I am not sure what is more shocking. That this is even an issue or that the current resolution is NAD .

With implicit constant initialization, if `i` was `const auto` instead of `auto`, if `foo` was a `constexpr` and if `reverse` was a `constexpr`, this example would not dangle and as such should not be ill-formed.

With C99 literal enclosing block lifetime, this example, AS IS, would not dangle.

In the identifying paper for this issue, Fix the range-based for loop, Rev1 ^[13], says the following:

“The Root Cause for the problem”

*“The reason for the undefined behavior above is that according to the current specification, the range-based for loop internally is **expanded to multiple statements**.”*

- *“First, we have some initializations using the for-range-initializer after the colon and”*
- *“Then, we are calling a low-level for loop”*

While certainly a factor, the problem is **NOT** that internally, the range-base for loop is expanded to multiple statements. It is rather that one of those statements has a scope of the statement instead of the scope of the containing block. The scoping difference between c99 and c++ rears it head again. From the programmers perspective, the issue in both cases is that c++ doesn't treat temporaries, unnamed variable as if they were named by the programmer just anonymously. The supposed correct usage highlights this fact.

```
auto v = foo();

for( auto i : reverse(v) ) { std::cout << i << std::endl; }
```

If you just name it, it works! Had reverse(foo()) been scoped to the block that contains the range based for loop than this too would have worked.

Should have worked	c99 would have worked	Progi
<pre>{// containing block for(auto i : reverse(foo())) { std::cout << i << std::endl; } }</pre>	<pre>{// containing block auto&& rg = reverse(foo()); auto pos = rg.begin(); auto end = rg.end(); for (; pos != end; ++pos) { int i = *pos; ... } }</pre>	<pre>{// a a f { } }</pre>

It should be no different had the programmer broken a compound statement into it's components and named them individually.

CWG1864 List-initialization of array objects

```
auto x[] = {1, 2, 3};
```

I am not exactly sure what is the problem here. With implicit constant initialization and if x was const auto instead of auto , this example would not dangle. With c99 literal enclosing block lifetime, this example, AS IS, would not dangle.

CWG2111 Array temporaries in reference binding

“Somewhat related to P2174 compound literals.”

That statement and the fact that all of the examples were `const` constant expressions means that issue is related to this proposal and the `NAD` may need to be revisited if this proposal ever gets accepted.

CWG914 Value-initialization of array types

I don't believe this issue is related to or would be benefited from this proposal.

Other languages

many native languages

Many native languages, including C and C++, already provide this capability by storing said instances in the `COFF String Table` of the portable executable format ^[14].

Java

Java automatically perform interning on its `String` class ^[15].

C# and other .NET languages

The .NET languages also performs interning on its `String` class ^[16]

many other languages

According to Wikipedia's article, `String interning` ^[17], Python, PHP, Lua, Ruby, Julia, Lisp, Scheme, Smalltalk and Objective-C's each has this capability in one fashion or another.

What is proposed here is increasing the interning that C++ already does, but is not limited to just strings, in order to reduce dangling references. The fact is, literals in `c++` is needless complicated with clearly unnecessary memory safety issues that impedes programmers coming to `c++` from other languages, even `c`.

Summary

The advantages to `c++` with this proposal is manifold.

- Reduce dangling
- Make `constexpr` literals less surprising for new and old developers alike
- Reduce the gap between `c++` and `c99` compound literals
- Improve the potential contribution of `c++`'s `constexpr` back to `c`

- Make string literals and `C++` literals more consistent with one another
- Reduce uninitialized and delayed initialization errors
- Taking a step closer to reducing undefined behavior in string literals
- Increase and improve upon the utilization of ROM and the benefits that entails

Frequently Asked Questions

What about locality of reference?

It is true that globals can be slower than locals because they are farther in memory from the code that uses them. So let me clarify, when I say `static storage duration`, I really mean **logically** `static storage duration`. If a type has a `constexpr` copy constructor or is a POD then there is nothing preventing the compiler from copying the global to a local that is closer to the executing code. Rather, the compiler **must** ensure that the code is always available; **effectively** `static storage duration`.

Consider this from an processor and assembly/machine language standpoint. A processor usually has instructions that works with memory. Whether that memory is ROM or is logically so because it is never written to by a program, then we have constants.

```
mov <register>,<memory>
```

A processor may also have specialized versions of common instructions where a constant value is taken as part of the instruction itself. This too is a constant. However, this constant is guaranteed closer to the code because it is physically a part of it.

```
mov <register>,<constant>
mov <memory>,<constant>
```

What is more interesting is these two examples of constants have different value categories since the ROM version is addressable and the instruction only version, clearly, is not. It should also be noted that the later unnamed/unaddressable version physically can't dangle.

References

- [illegible]

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